

Polynomial-time decision of local-unitary equivalence for graph states

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Abstract

We give a deterministic polynomial-time algorithm for deciding local-unitary equivalence of labelled graph states. The starting point is the quasi-polynomial algorithm of Claudet and Perdrix, whose expensive step lists high-order incidence constraints for generalized local complementations. We show that the same row module over $\mathbb{Z}/2^r\mathbb{Z}$ can be generated from pair and triple rows by a polynomial number of diagonal-map and Hermite-normal-form operations. The closure takes at most $\max\{0, r - 2\}$ enlargement steps. An integer dual lattice then gives the complete modular kernel and its binary space of edge toggles, together with witnessing multiplicities. Substituting this computation into the published graph reduction gives polynomial bit complexity. We prove that the auxiliary graphs satisfy the constrained local-Clifford algorithm's hypotheses and handle disconnected inputs and zero toggle spaces separately.

1 Introduction

A graph state associates an n -qubit pure state with a finite simple graph. Its graph representation makes local-Clifford transformations amenable to discrete methods, and local-Clifford equivalence can be decided efficiently [1, 2]. Local-unitary equivalence allows arbitrary single-qubit unitaries and is a different decision problem: LU-equivalent graph states need not be LC-equivalent [3]. Throughout this paper the vertices are labelled: the algorithm does not optimize over qubit permutations.

Claudet and Perdrix [4] reduced local-unitary equivalence to a sequence of graph and linear-algebra computations and obtained the bound $n^{\log_2 n + O(1)}$. Their standard-form reduction leaves one generalized local complementation, together with ordinary local complementations on a specified set of vertices. The generalized operation is described by linear congruences, but the direct description has a row for every relevant high-order subset. At level r , this creates $O(n^{r+1})$ rows. The level required for the general local-unitary question is logarithmic in n . Whether the general decision problem admits a polynomial-time algorithm is also explicitly posed in Claudet's thesis [7, Chapter 8, p. 107].

The subset enumeration can be avoided. Except at the pair-to-triple transition, adjoining a vertex to a subset acts on its weighted incidence row by a fixed diagonal linear map. Starting with both pair and triple rows therefore suffices. Repeatedly applying these maps, and reducing the row lattice to Hermite form, computes all constraints without retaining the subsets that index them. Only $r - 2$ enlargement steps can be needed when $r \geq 2$.

Theorem 1.1 (Main result). *There is a deterministic classical algorithm that, given two finite simple graphs G and H on the same labelled vertex set of size $n \geq 1$, decides in polynomial bit complexity whether their graph states are local-unitary equivalent, up to global phase. No connectivity, degree, or graph-class promise is required.*

The graph-theoretic reductions are due to Claudet and Perdrix. The new ingredient is the polynomial computation of the incidence module; the complete toggle space then follows by modular linear algebra. Sections 3 and 4 give these constructions. Sections 5 and 6 explain their use in the published reduction, including the hypotheses needed for its constrained local-Clifford step. Section 7 reports exact diagnostic checks and their limits.

2 Problem and notation

For a finite simple graph $G = (V, E)$ with $V = [n]$, define

$$|G\rangle = \prod_{\{u,v\} \in E} \text{CZ}_{uv} |+\rangle^{\otimes n}, \quad |+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}. \quad (1)$$

Local-unitary equivalence means that

$$|H\rangle = e^{i\phi} \left(\bigotimes_{v=1}^n U_v \right) |G\rangle \quad \text{for some } U_v \in U(2), \phi \in \mathbb{R}. \quad (2)$$

The labels in (2) are fixed. A local complementation at v toggles every edge among distinct neighbors of v . We write $G \star v$ for the resulting graph and use LC for equivalence under sequences of these operations.

For a multiset S of vertices, $s(x)$ denotes its nonnegative integer multiplicity and $\text{supp}(S) = \{x : s(x) > 0\}$. For $K \subseteq V$, let

$$c_S(K) = \sum_{x: K \subseteq N_G(x)} s(x) \quad (3)$$

be its common-neighbor multiplicity. The following conventions are those of [4, Definitions 1–2].

Definition 2.1. Let $r \geq 1$. An independent multiset S is r -incident if, for every $0 \leq j < r$ and every $K \subseteq V \setminus \text{supp}(S)$ with $|K| = j + 2$,

$$c_S(K) \equiv 0 \pmod{2^{r-j-\delta_{j0}}}. \quad (4)$$

Its r -local complementation toggles a pair outside $\text{supp}(S)$ precisely when its common-neighbor multiplicity is 2^{r-1} modulo 2^r . Pairs meeting the support are unchanged.

Let LC_r denote the local-equivalence relation generated by single-qubit Hadamards and $Z(\pi/2^r)$ rotations, in the convention of [4]. Thus $\text{LC}_1 = \text{LC}$, and every r -local complementation is a local-unitary transformation. The equivalence between the graph operations and these unitary relations comes from the graphical characterization of Claudet and Perdrix [5]; the conventions used here are summarized in [4, Section 2.3].

Our input graphs have adjacency-matrix encodings. Arithmetic complexity means bit complexity. The auxiliary parameter r is charged according to its value, not merely its binary encoding length. This is sufficient because the final algorithm chooses $r = O(\log n)$. The ring $R = \mathbb{Z}/q\mathbb{Z}$, where $q = 2^r$, is not a field when $r > 1$; its kernels need not be free modules.

3 Compression of the incidence module

We first state the algebraic result independently of graph standard forms. Let $a_1, \dots, a_t \in \{0, 1\}^m$. For $K \subseteq [t]$, define the coordinatewise product

$$a_K = \bigodot_{v \in K} a_v, \quad w(k) = \begin{cases} 1, & k = 2, \\ k - 2, & k \geq 3. \end{cases} \quad (5)$$

The weighted incidence module is

$$M_r = \text{span}_R \{2^{w(|K|)} a_K : 2 \leq |K| \leq r + 1\} \subseteq R^m. \quad (6)$$

Subsets larger than t contribute no rows.

Theorem 3.1 (Polynomial incidence compression). *A generating matrix for M_r with at most m rows can be constructed deterministically in time polynomial in m, t, r . The construction uses at most*

$$B(r, t) = \max\{0, \min\{r - 2, t - 3\}\} \quad (7)$$

strict lattice enlargements, and its retained integer basis entries have $O(r + 1)$ bits.

3.1 The closure identity

For each $v \in [t]$, define the R -linear map

$$T_v(z) = 2a_v \odot z. \quad (8)$$

Let N_r be the smallest submodule containing $2a_K$ for $|K| = 2$ or 3 and invariant under all the maps T_v .

Lemma 3.2. $M_r = N_r$.

Proof. The pair seeds occur in (6). The triple seeds occur when $r \geq 2$; when $r = 1$, all seeds are zero modulo two. Thus every seed belongs to M_r .

We check invariance on a generating row $2^{w(k)} a_K$, where $k = |K|$. If $v \in K$, the binary coordinates satisfy $a_v \odot a_K = a_K$, so T_v gives twice that row. If $v \notin K$ and $k = 2$, the result is $4a_{K \cup \{v\}}$, twice the triple seed. For $r = 1$ this result is zero. If $v \notin K$ and $k \geq 3$, then

$$T_v(2^{k-2} a_K) = 2^{k-1} a_{K \cup \{v\}}, \quad (9)$$

which is the next prescribed row, unless $k = r + 1$, in which case it vanishes modulo 2^r . Hence M_r is invariant and $N_r \subseteq M_r$.

Conversely, if $|K| = k \geq 4$, choose three vertices of K and start with their triple seed. Applying T_v for the remaining $k - 3$ distinct vertices gives coefficient $2 \cdot 2^{k-3} = 2^{k-2}$. Every generator in (6) is therefore in N_r . Pair and triple generators were already seeds. This proves equality; in particular, repeated application of a vertex introduces no extraneous constraints. $\square$

3.2 A bounded lattice computation

We fix the following row-Hermite convention. For a full-rank lattice $L \subseteq \mathbb{Z}^m$, its basis matrix H is lower triangular, with

$$H_{ii} > 0, \quad H_{ij} = 0 \ (j > i), \quad 0 \leq H_{ij} < H_{jj} \ (j < i). \quad (10)$$

The rows generate L over $\mathbb{Z}$. Transposing a column-Hermite computation gives this convention.

Lemma 3.3 (Size and lifting of a lattice basis). *Suppose $q\mathbb{Z}^m \subseteq L \subseteq \mathbb{Z}^m$, and let H be its row-Hermite basis. Each H_{ii} divides q , and every entry of H lies between zero and q . If $\pi : \mathbb{Z}^m \rightarrow (\mathbb{Z}/q\mathbb{Z})^m$ is reduction modulo q , then $L = \pi^{-1}(\pi(L))$.*

Proof. For each i , write $qe_i = cH$ with $c \in \mathbb{Z}^m$, since $qe_i \in L$. Compare columns from m down to $i + 1$. Lower triangularity and the nonzero pivots successively give $c_j = 0$ for all $j > i$. Column i then gives $c_i H_{ii} = q$, proving $H_{ii} \mid q$. The off-diagonal bounds follow from (10).

If $z \in \pi^{-1}(\pi(L))$, choose $y \in L$ with $z - y \in q\mathbb{Z}^m$. Both y and $z - y$ lie in L , so $z \in L$. The other inclusion is immediate. In particular, two lattices containing $q\mathbb{Z}^m$ are equal whenever their images modulo q are equal. $\square$

Proof of Theorem 3.1. The case $m = 0$ is immediate, so assume $m > 0$. Lift the seed rows to integers and let $L_0 \subseteq \mathbb{Z}^m$ be the lattice generated by those rows and $q\mathbb{Z}^m$. Extend (8) to the same integer diagonal map and iterate

$$L_{i+1} = L_i + \sum_{v=1}^t T_v(L_i). \quad (11)$$

After each step, retain a full-rank row Hermite basis. Applying T_v to a basis suffices to generate $T_v(L_i)$.

The image of every L_i modulo q is contained in M_r , by the invariance proved in Lemma 3.2. Conversely, a prescribed row indexed by a set of size $k \geq 4$ is obtained from a triple seed after $k - 3$ applications of the maps. The largest relevant size is $\min\{t, r + 1\}$. Thus every prescribed row is present by step $B(r, t)$; when $r \leq 2$ or $t \leq 3$, the seeds already suffice. Every L_i contains $q\mathbb{Z}^m$, and $T_v(q\mathbb{Z}^m) \subseteq q\mathbb{Z}^m$. Lemma 3.3 therefore makes equality of their modular images equivalent to equality of the integer lattices. Hence $L_{B(r,t)}$ is stable. If equality occurs earlier, invariance prevents any later change. A further Hermite computation detects stability.

The initial matrix has at most $m + \binom{t}{2} + \binom{t}{3}$ rows and m columns. Every subsequent matrix has at most $m(1 + t)$ rows. By Lemma 3.3, every retained entry lies in $[0, q]$ and has $O(r + 1)$ bits. Applying T_v gives entries of magnitude at most $2q$. There are at most $B(r, t) + 2$ Hermite computations: the initial reduction, at most $B(r, t)$ strict enlargements, and a final stability check.

Deterministic Hermite-normal-form algorithms have polynomial bit complexity in their matrix dimensions and input bit lengths [8, Theorem 2 and the rectangular extension $H(m, n, A)$, p. 504]. Apply that full-row-rank rectangular algorithm to the transpose of our generating matrix, which has rank m because it contains qI_m . Polynomially many calls on the bounded matrices above establish the asserted bit complexity, including intermediate arithmetic inside each call. Reducing the final m basis rows modulo q gives a generating matrix B for M_r . Zero rows may be omitted. These are module generators; they need not form a free basis over R . $\square$

Algorithm 1: weighted incidence compression

1. Form the pair and triple rows $2a_K$ and the rows of qI_m .
2. Replace their integer row span by a canonical Hermite basis H .
3. Compute a Hermite basis H' of the rows of H and all $T_v(H)$.
4. If $H' = H$, return $H \bmod q$; otherwise set $H \leftarrow H'$ and repeat step 3.

For $m = 0$, return the empty matrix immediately.

Corollary 3.4. *For $s \in R^m$, the exponentially indexed conditions*

$$2^{w(|K|)} a_K \cdot s = 0 \pmod{q} \quad (2 \leq |K| \leq r + 1) \quad (12)$$

hold if and only if $Bs = 0$ in R .

Proof. Annihilating generators of a row module is equivalent to annihilating the entire module. Apply Theorem 3.1. $\square$

4 The complete toggle space and its witnesses

Fix a graph G , an independent set $X \subseteq V$, and $Y = V \setminus X$. Put $m = |X|$, $q = 2^r$, and $h = 2^{r-1}$. For each $v \in Y$, let $a_v \in \{0, 1\}^X$ be its adjacency vector into X . Let B be the compressed matrix and

$$\Sigma = \ker_R B. \quad (13)$$

4.1 Matching the precise support convention

Lemma 4.1. *The vectors in Σ , represented by multiplicities in $\{0, \dots, q-1\}$ on X , are exactly the r -incident multisets supported in X .*

Proof. For $K \subseteq Y$, the common-neighbor multiplicity is $a_K \cdot s$. When $k = |K|$, condition (4) requires divisibility by $2^{r-w(k)}$, exactly equivalent to (12). This equivalence also holds for $r = 1$, when the pair condition is vacuous.

The actual support of a multiplicity vector may be smaller than X . If a set $K \subseteq V \setminus \text{supp}(S)$ contains some $x \in X$, no vertex of X is adjacent to every vertex of K : independence of X forbids adjacency to x . Its common-neighbor multiplicity in S is therefore zero. All such extra conditions in Definition 2.1 hold automatically. These observations establish both directions. Reducing multiplicities modulo q preserves all relevant divisibilities. $\square$

For every unordered pair $p \in \binom{Y}{2}$, define

$$f(s)_p = \frac{a_p \cdot \tilde{s}}{h} \pmod{2}, \quad s \in \Sigma, \quad (14)$$

where $\tilde{s}$ is any integer lift. Let $\Omega = f(\Sigma)$.

Lemma 4.2. *The map in (14) is well-defined and satisfies*

$$f(s+t) = f(s) + f(t), \quad f(cs) = (c \bmod 2)f(s). \quad (15)$$

Consequently Ω is a binary linear subspace.

Proof. The pair constraint is $2a_p \cdot s = 0 \pmod{q}$, so every integer lift has $a_p \cdot \tilde{s}$ divisible by h . For $r = 1$, this is automatic because $h = 1$. Changing the lift by qz changes the quotient by $2a_p \cdot z$, an even integer. The map is therefore well-defined.

If $\widetilde{s+t} = \tilde{s} + \tilde{t} - qz$, then

$$\frac{a_p \cdot \widetilde{s+t}}{h} = \frac{a_p \cdot \tilde{s}}{h} + \frac{a_p \cdot \tilde{t}}{h} - 2a_p \cdot z. \quad (16)$$

Reducing modulo two removes the last term. For a lift $\tilde{c}$ of a scalar $c \in R$, writing $\tilde{c}s = \tilde{c}\tilde{s} - qz$ gives the scalar identity in the same way. Reduction $R \rightarrow \mathbb{F}_2$ is well-defined because q is even, establishing (15). $\square$

4.2 Generating a possibly nonfree kernel

Lemma 4.3. *Given a $b \times m$ matrix B over R , with $b \leq m$, a generating set of $\ker_R B$ with at most m elements can be computed deterministically in polynomial bit complexity in m, r .*

Proof. The case $m = 0$ is immediate. Otherwise, let H be a full-rank row Hermite basis of the integer lattice generated by a lift of B and qI_m . The conditions $Bs = 0 \pmod{q}$ and $Hs = 0 \pmod{q}$ are equivalent. Because the row lattice of H contains $q\mathbb{Z}^m$, there is an integer matrix C with $qI_m = CH$. It follows that

$$K = qH^{-1} = C \quad (17)$$

is an integer matrix.

The columns of K , reduced modulo q , generate the kernel. Each column is a solution because $HK = qI_m$. Conversely, for any integer lift of a solution s , write $Hs = qz$ with $z \in \mathbb{Z}^m$. Then $s = Kz$, proving completeness. The generators need not be independent over R ; no freeness assumption is made.

Computing H has polynomial bit complexity, and Lemma 3.3 bounds its entries by q . There is no need to compute a rational inverse. Solve $HK = qI_m$ by forward substitution, column by column:

$$k_{ij} = \frac{q\mathbf{1}_{i=j} - \sum_{\ell < i} H_{i\ell} k_{\ell j}}{H_{ii}}, \quad i = 1, \dots, m. \quad (18)$$

Every division is exact, because (17) already proves that the unique solution is integral. For a fixed column j , the entries above j vanish and $|k_{jj}| \leq q$. If $S_i = \sum_{\ell \leq i} |k_{\ell j}|$, then $S_i \leq (1+q)S_{i-1}$ for $i > j$. Hence

$$|k_{ij}| \leq q(1+q)^{m-1}. \quad (19)$$

All entries and intermediate sums have $O(m(r+1) + \log(m+1))$ bits. Equation (18) uses $O(m^3)$ integer arithmetic operations on numbers of this size, so even elementary integer algorithms suffice for polynomial bit complexity. Finally reduce the columns modulo q . If B came from Algorithm 1, reuse its already computed integer basis H .

For example, $B = (2)$ over $\mathbb{Z}/4\mathbb{Z}$ gives $H = (2)$ and $K = (2)$. Its kernel is $\{0, 2\}$, generated by an element annihilated by two. This illustrates why treating R as a field, or assuming a free kernel, would be incorrect. $\square$

Theorem 4.4 (Complete image with preimages). *In polynomial bit complexity in n, r , one can compute a basis $\omega_1, \dots, \omega_d$ of Ω , with $d \leq |X|$, and vectors $s_i \in \Sigma$ satisfying $f(s_i) = \omega_i$. Membership in Ω is decidable in polynomial time, with a witnessing multiplicity vector for every member.*

Proof. Apply Lemma 4.3 to obtain generators $g_1, \dots, g_\ell$ of Σ , with $\ell \leq m$. For each g_j and pair p , compute the residue $c_p = a_p \cdot g_j \pmod{q}$ in $\{0, \dots, q-1\}$. It belongs to $\{0, h\}$ by the pair constraint. Thus $f(g_j)_p = c_p/h$ is an exact integer division; it does not divide by a nonunit in R .

Every $s \in \Sigma$ is an R -linear combination $\sum_j c_j g_j$. By Lemma 4.2,

$$f(s) = \sum_j (c_j \pmod{2}) f(g_j). \quad (20)$$

Conversely, every binary combination of the images is the image of the corresponding sum of the generators. Hence these at most m images span all of Ω .

Perform binary Gaussian elimination on them, carrying along their preimages over R . Whenever image rows are added, add the corresponding preimages modulo q . The invariant

$f(s_i) = \omega_i$ survives by (15). Remove dependent zero image rows. If $\delta \in \Omega$, solve $\delta = \sum_i \epsilon_i \omega_i$ over $\mathbb{F}_2$ and return $s = \sum_i \epsilon_i s_i \bmod q$.

There are $\binom{|Y|}{2} = O(n^2)$ image coordinates and at most m generators. Image computation uses $O(n^2 m^2)$ elementary arithmetic operations. Before reducing a dot product modulo q , its value is at most $m(q-1)$, requiring $O(r + \log(m+1))$ bits. Binary elimination uses $O(n^2 m^2)$ field operations, and at most $O(m^3)$ additions of r -bit witness coordinates. Thus the carried witnesses do not hide a growing precision requirement. Combined with the compression and kernel bounds, this proves polynomial bit complexity. An equation description of Ω may alternatively be obtained from a basis of its binary orthogonal complement. $\square$

Corollary 4.5 (One prescribed-support operation). *A graph H is obtained from G by one valid r -local complementation supported in X if and only if the two graphs agree on pairs meeting X and their edge difference on Y belongs to Ω . The condition is decidable in polynomial bit complexity in n, r , and an accepting instance has constructible binary-encoded multiplicities.*

Proof. Lemma 4.1 identifies all valid multiplicities. For a pair in Y , the toggle predicate is exactly (14). A pair meeting the actual support is unchanged by definition. A pair meeting X outside that support has no common neighbor in X , so its common-neighbor multiplicity is zero and it is unchanged as well. These facts prove necessity and sufficiency. Representatives below q have at most r bits each. $\square$

5 The published graph reduction

All numbered citations to Claudet and Perdrix [4] below refer to the *PDF* of arXiv:2502.06566v3. Its HTML rendering uses different numbers; Appendix A records the correspondence. We state the imported results separately from the algebra proved above.

5.1 Local sets and standard forms

Write Γ for the adjacency matrix over $\mathbb{F}_2$, and X_u, Z_u for single-qubit Paulis. The graph-state stabilizer has generators $X_u \prod_{v \in N(u)} Z_v$. For a vertex set D with indicator d , their product has support $D \cup \text{Odd}_G(D)$; at vertex u its Pauli, up to phase, is specified by $(d_u, (\Gamma d)_u)$. The three nonzero pairs $(1, 0), (1, 1), (0, 1)$ represent X, Y, Z , respectively.

A local set is a nonempty support of this form. An MLS cover $\mathcal{M}$ is a family of inclusion-minimal local sets covering every vertex. For a vertex u , consider every generator D of every member of $\mathcal{M}$ containing u . Its type is X, Y , or Z if all these Pauli pairs equal that same nonzero pair, and $\perp$ otherwise [4, Definitions 12–13, pp. 8–9]. A cover can be computed in polynomial time [6]; the algorithm does not enumerate all minimal local sets.

Standard form has three requirements [4, Definition 14, p. 9]: no vertex has type Y ; every neighbor v of a type- X vertex u has type Z and satisfies $u \prec v$ in the fixed vertex order; and each star $\{u\} \cup N(u)$ at a type- X vertex belongs to $\mathcal{M}$. In particular, the type- X set is independent.

The common-standard-form procedure [4, Lemma 16 and Appendix C] either rejects LU-equivalence soundly or returns ordinary-LC-equivalent representatives in standard form with respect to a common MLS cover, in $O(n^{6.38})$ time. For LU-equivalent representatives, corresponding types and type- X neighborhoods agree [4, Lemma 17]. The generalized-complementation characterization then gives the following precise interface: surviving standard forms are LC_r -equivalent if and only if one r -local complementation supported on their type- X set, together with ordinary complementations on type- $\perp$ vertices, transforms one into the other [4, Lemma 18]. The reverse implication follows because each permitted operation is an LC_r operation.

Write X, Z, W for the type- X , type- Z , and type- $\perp$ sets. Standard form gives $N(X) \subseteq Z$. Ordinary complementation at $w \in W$ changes only edges among $N(w)$, which is disjoint from X . It therefore preserves all neighborhoods of X . A generalized operation on X changes only Z - Z edges, preserving $N(w)$. Each operation thus preserves the data specifying the other's toggles, so they commute. The same reasoning applies throughout a sequence on W : no such step creates an edge from W to X .

5.2 Where the compression enters

Lemma 19 of [4] describes the full space Ω of changes on Z - Z edges by enumerating the incidence constraints. Their coefficients are precisely $2^{w(|K|)}a_K$ from (6). Theorems 3.1 and 4.4 compute the same space and witnessing multiplicities in polynomial bit complexity. The binary linearity of Ω is already present in the source; the improvement here is its computation.

Given a basis of Ω , the construction preceding Lemma 20 of [4] deletes X and adds degree-two auxiliary vertices for its nonzero coordinates. Lemma 20 and Appendix D reduce the remaining problem to restricted ordinary LC. Lemma 21 encodes the restrictions as linear equations on local-Clifford variables. We give the construction and its correctness in Section 6, including the binary equations and the algorithm for satisfying their determinant conditions. The latter uses the affine-structure theorem in [4, Lemma 8], extending Bouchet's algorithm [1].

Finally, [4, Corollary 29, p. 14] gives $LU \Rightarrow LC_r$ on graphs with at most $2^{r+3} - 1$ vertices. Since every LC_r operation is a local unitary, the two decision problems coincide within that size bound. These standardization, characterization, and affine-structure results are the external mathematical dependencies of the proof.

6 The full algorithm

6.1 Components and labels

Lemma 6.1. *LU-equivalent labelled graph states have the same partition of their vertex set into connected components. Equivalence then holds if and only if it holds on each corresponding component.*

Proof. A graph state factors across a labelled cut $A : V \setminus A$ if and only if no edge crosses that cut. To see the nontrivial direction, remove the diagonal phases contributed by edges lying entirely within either side of the cut. Up to a common normalization, the bipartite coefficient matrix becomes

$$C_{x,y} = (-1)^{x^\top \Gamma_{A,V \setminus A} y}. \quad (21)$$

If a crossing entry of Γ equals one, the rows indexed by the zero vector and the corresponding single-bit vector, and the analogous two columns, give the minor

$$\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}. \quad (22)$$

It has nonzero determinant. The state therefore has Schmidt rank at least two across the cut. With no crossing edges, the coefficient matrix factors.

Product unitaries preserve factorization across every fixed labelled cut. The finest partition into graph-state product factors is thus the connected-component partition, which must be the same for both inputs. When partitions coincide, a global product-unitary equality implies equality of the pure-state factors up to component phases, for example by taking reduced states on each component. Conversely, tensoring component transformations supplies a transformation on the whole graph, with their phases absorbed into the global phase. $\square$

Compute these partitions and reject if they differ. One-vertex components are immediate. On a common connected component of order $v \geq 2$, choose

$$r = \max\{1, \lceil \log_2(v+1) \rceil\}. \quad (23)$$

This gives $v \leq 2^{r+3} - 1$ and $r = O(\log v)$, so Corollary 29 of [4] makes the LU and LC_r questions identical.

Apply the common-standard-form construction and the type and neighborhood tests from [4, Lemmas 16–17]. If they succeed, let G_1, G_2 be the standard forms and let X, Z, W be their common type- X , type- Z , and type- $\perp$ sets. The graphs remain connected: any edge removed by ordinary complementation at a vertex is replaced by a two-edge path through that vertex. Compute the image basis from Theorem 4.4 using G_1 and X . Its only possibly nonzero coordinates are pairs in Z , since vertices outside $X \cup Z$ have no neighbor in X . Restricting to these coordinates loses no information.

6.2 The nonzero-image branch

Suppose $d = \dim \Omega > 0$. For each basis vector ω_i and each pair $\{u, w\} \subseteq Z$ on which it is one, add a new vertex p_{uw}^i joined only to u, w , after deleting X . Use the same new labels in both graphs. Let P_i be the set of new vertices for ω_i . Their total number is at most $d \binom{|Z|}{2}$, and

$$N \leq v + d \binom{v}{2} \leq v + |X| \binom{v}{2} = O(v^3). \quad (24)$$

Lemma 6.2 (Applicability of constrained LC). *When $d > 0$, both augmented graphs are connected and have an even-degree vertex.*

Proof. Every basis vector is nonzero, so at least one auxiliary vertex is added; each such vertex has degree two.

For connectivity, fix $x \in X$. The multiplicity vector $h\mathbf{e}_x$ is in Σ , because every weighted incidence row has an even coefficient and hence its scalar product with this vector is divisible by q . Its toggle vector is exactly the clique on $N_{G_1}(x)$. As the basis vectors span Ω , every pair of distinct neighbors of x must occur in at least one basis vector. Some auxiliary vertex therefore supplies a path of length two between each such pair.

Take any two vertices outside X and a simple path between them in G_1 . Independence of X ensures that each passage through an internal vertex $x \in X$ has two neighbors outside X . Replace that passage by the auxiliary path just described. Vertices of X with degree at most one cannot occur internally on the original simple path. Thus the remainder together with the auxiliary vertices is connected. The remainder is nonempty because G_1 is connected, has at least two vertices, and X is independent. All new vertices attach to that remainder. The same argument applies to G_2 , using its connectedness and the identical neighborhoods on X . $\square$

Lemma 6.3 (Restricted graph operations). *The original standard forms are LC_r -equivalent if and only if their augmented graphs are related by ordinary local complementations on $W \cup \bigcup_i P_i$, with every group P_i either used in its entirety, once per vertex, or not used.*

Proof. This is the reduction of [4, Lemma 20 and Appendix D]; we spell out why a complete image basis suffices. The auxiliary vertices form an independent set, have neighbors only in Z , and have no neighbors in W . Complementing p_{uw}^i toggles only the edge $\{u, w\}$. Using all of P_i therefore adds ω_i to the edge vector on Z .

A complementation at $w \in W$ preserves every auxiliary neighborhood, since none contains w . Complementation at an auxiliary vertex preserves $N(w)$, since it changes only Z – Z edges.

The operations commute, and these properties persist after each allowed step. Thus an allowed sequence can be ordered with all auxiliary operations first, followed by operations on W .

If the standard forms are LC_r -equivalent, their transformation has a toggle $\omega \in \Omega$ followed by operations on W . Express $\omega = \sum_i \epsilon_i \omega_i$ and use exactly the groups with $\epsilon_i = 1$. After deleting X , this performs the same toggle; the remaining operations on W have the same effect. Conversely, an allowed augmented sequence selects a binary sum $\omega \in \Omega$. Theorem 4.4 supplies a valid multiplicity vector with this image. Restore X , perform its generalized operation, and then the same sequence on W . The original neighborhoods of X are unchanged throughout, so the result is exactly the second standard form. $\square$

6.3 The binary constrained-Clifford decision

Let Γ, Γ' be the augmented adjacency matrices, now of order N . Introduce $4N$ binary variables $\alpha_u, \beta_u, \gamma_u, \delta_u$ and their diagonal matrices $\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}$. The local-Clifford characterization [4, Proposition 7, pp. 6–7; Appendix A], originating in [1], is

$$\Gamma \mathbf{B} \Gamma' + \Gamma \mathbf{A} + \mathbf{D} \Gamma' + \mathbf{C} = 0, \quad (25)$$

$$\alpha_u \delta_u + \beta_u \gamma_u = 1 \quad (u \in V^\#). \quad (26)$$

Here $V^\#$ is the augmented vertex set, and all operations are in $\mathbb{F}_2$. The first equation is linear in the $4N$ variables: its (u, v) entry is

$$\sum_w \Gamma_{uw} \beta_w \Gamma'_{wv} + \Gamma_{uv} \alpha_v + \delta_u \Gamma'_{uv} + \mathbf{1}_{u=v} \gamma_u = 0. \quad (27)$$

The determinant conditions (26) are quadratic and cannot simply be omitted after Gaussian elimination.

The additional homogeneous linear equations for Lemma 6.3 are those of [4, Lemma 21, pp. 11–12]:

$$\beta_u = 0 \quad (u \in Z), \quad \gamma_p = 0 \quad (p \in \bigcup_i P_i), \quad \beta_p = \beta_{p_i} \quad (p \in P_i \setminus \{p_i\}), \quad (28)$$

where p_i is one representative of each nonempty group. If $P = \bigcup_i P_i$, there are $\ell = |Z| + 2|P| - d \leq 2N$ such equations.

For completeness, the role of these constraints can be seen in the solution-to-sequence conversion of [4, Appendix A]. At an auxiliary vertex, $\gamma_p = 0$ and (26) force $\alpha_p = \delta_p = 1$. If $\beta_p = 1$, complementing p on the source graph replaces β_p by $\beta_p + \delta_p = 0$, and leaves the other β coordinates unchanged. The auxiliary vertices are independent, so these operations may be performed first for all groups with $\beta = 1$. They leave identity local coordinates on the auxiliary vertices. The remaining solution has $\beta = 0$ outside W .

Here are the coordinate updates underlying that extraction. Complementing w on the source graph replaces its adjacency matrix by that of $G \star w$, and replaces the solution by

$$\begin{aligned} \alpha'_u &= \alpha_u + \mathbf{1}_{u=w} \gamma_w, & \beta'_u &= \beta_u + \mathbf{1}_{u=w} \delta_w, \\ \gamma'_u &= \gamma_u + \Gamma_{uw} \alpha_u, & \delta'_u &= \delta_u + \Gamma_{uw} \beta_u. \end{aligned} \quad (29)$$

These identities preserve (25)–(26) [4, Appendix A, pp. 18–20]. If some w has $\beta_w = \delta_w = 1$, this step removes w from the support of β and changes no other β coordinate. If β is nonzero but no such w exists, every vertex u in its support has $\delta_u = 0$ and $\gamma_u = 1$. The (u, u) entry of (25) then gives

$$\sum_w \Gamma_{uw} \beta_w \Gamma'_{wu} = 1. \quad (30)$$

Thus some neighbor w of u also has $\beta_w = 1$ and $\delta_w = 0$. Applying the three complementations u, w, u removes both u and w from the support of β , as follows by applying (29) three times; other β coordinates do not change. Repeating these reductions terminates with $\beta = 0$. The determinant equations then force $\alpha = \delta = 1$, and the diagonal part of (25) forces $\gamma = 0$. The remaining equation is $\Gamma = \Gamma'$, proving that the extracted sequence reaches the target.

Since every step above uses only vertices in the current support of β , the sequence remaining after the auxiliary steps is supported on W . No operation there changes an auxiliary neighborhood. Conversely, the same update identities, applied in reverse to a sequence of Lemma 6.3, produce a solution of (25)–(28). An unused auxiliary has $\beta = \gamma = 0$; one used once has $\beta = 1, \gamma = 0$; and β stays zero on Z . This proves that the binary constraints encode the required graph predicate, as asserted in the cited lemma.

Lemma 6.4 (Testing the determinant conditions). *On two connected graphs of order N , each with an even-degree vertex, equations (25)–(26) together with ℓ homogeneous linear constraints can be decided in $O((N^2 + \ell)N^2)$ binary operations.*

Proof. Compute a basis $b_1, \dots, b_k$ of the linear solution space S , with $k \leq 4N$, by Gaussian elimination. This costs $O((N^2 + \ell)N^2)$ operations. If $k \leq 4$, test all at most 16 vectors against (26).

For $k > 4$, the affine-structure result [4, Lemma 8, p. 7] states that the set $C \subseteq S$ additionally satisfying the determinant equations is empty or is an affine subspace of codimension at most two. Suppose $C = a + U$ is nonempty, with $\dim(S/U) = c \leq 2$. The images of the b_i span S/U ; select c of them as a quotient basis. The coset $a + U$ is a sum of at most c of those basis images. Hence one of

$$0, \quad b_i, \quad b_i + b_j \quad (i < j) \tag{31}$$

belongs to C . Test this list, of length $O(N^2)$, at a cost $O(N)$ per candidate. If none passes, C is empty. This proves the bound and makes explicit the finite search in [4, Proposition 10]. $\square$

Lemma 6.2 establishes connectedness and the even-degree condition for both augmented graphs. Lemmas 6.3 and 6.4, with (28), therefore decide the required predicate in polynomial time in their actual order N .

6.4 The zero-image branch

Lemma 6.5. *If $\Omega = \{0\}$, then G_1 and G_2 are LC_r -equivalent if and only if they are ordinarily LC -equivalent.*

Proof. One implication is immediate. For the other, [4, Lemma 18] expresses a LC_r equivalence by ordinary complementations on W and one valid generalized operation on X . These operations commute. Since $\Omega = \{0\}$, the latter operation changes no edge of G_1 and can be omitted. Only ordinary complementations remain. $\square$

In this case run the ordinary LC algorithm [2] on the original standard forms G_1, G_2 . Deleting X first could leave disconnected graphs or graphs without an even-degree vertex, outside the constrained routine's hypotheses. Lemma 6.5 supplies a separate exact decision rule, including the case $X = \emptyset$.

6.5 Correctness and bit complexity

Algorithm 2: labelled graph-state LU equivalence

1. Compare labelled connected-component partitions. Reject if they differ.
2. For each common component of order $v \geq 2$, choose r by (23).
3. Use [4, Lemmas 16–17] to standardize and check types and type- X neighborhoods. Reject on failure.
4. Use Algorithm 1 and Theorem 4.4 to compute the complete toggle basis.
5. If $d = 0$, decide ordinary LC on the original standard forms. If $d > 0$, build the matched gadgets and decide constrained LC using (28).
6. Accept exactly when every component test accepts; singletons always accept.

Proof of Theorem 1.1. Lemma 6.1 justifies the first step and all componentwise decisions. The cited standard-form lemmas make every standardization or type-based rejection sound. On a surviving component, Lemma 4.1 and Theorem 4.4 compute precisely the full toggle space used in [4, Lemma 20]. For $d > 0$, Lemmas 6.2–6.4 make the constrained test equivalent to LC_r equivalence. For $d = 0$, Lemma 6.5 gives that equivalence directly. Corollary 29 of [4] and (23) identify it with LU-equivalence. This proves soundness and completeness, and every construction preserves the original labels.

For the complexity bound, the common-cover stage takes $O(v^{6.38})$ time. Compression uses at most $B(r, |V \setminus X|)$ strict enlargements, each represented by a matrix with polynomially many rows and $O(r + 1)$ -bit entries. Its Hermite computations therefore have polynomial bit complexity. Kernel construction, exact image evaluation, and binary elimination have polynomial matrix sizes and bounded residue lengths. Since $r = O(\log v)$ and $2^r \leq 2(v + 1)$, no hidden exponential precision parameter is introduced.

The augmented graphs have $N = O(v^3)$ vertices and $\ell \leq 2N$ extra constraints. Lemma 6.4 therefore bounds this binary decision stage by $O(N^4) = O(v^{12})$. This bound concerns the augmented-graph stage; the algebraic stages have their own polynomial bit bounds, and we do not assert that v^{12} bounds the entire implementation. The zero-image branch uses an ordinary polynomial LC test. Summing a fixed polynomial bound over at most n components remains polynomial in n . We do not optimize the exponent or claim a practical benchmark for the complete procedure. $\square$

7 Exact diagnostics and reproducibility

The accompanying code tests the algebraic constructions and parts of the graph reduction using exact arithmetic. It does not implement the polynomial MLS standardization algorithm, and the finite checks do not prove Theorem 1.1.

7.1 Algebraic checks

The original scripts compare closure with direct subset enumeration on 45 systems, and kernels and images with exhaustive multiplicity enumeration on 36 systems. All passed. However, every original closure example stabilized at its seed lattice. A second script addresses this gap using all Boolean column patterns on $t \in \{3, 4, 5, 6\}$ rows, at six levels. Twelve of its 24 systems require strict enlargement; retaining the seeds alone fails in each of those twelve. The script

also checks all 2×2 matrices over $\mathbb{Z}/4\mathbb{Z}$, 384 binary incidence systems, and five boundary cases. These records are preserved in the archive.

A fresh implementation uses the dual-lattice formula (17); the earlier implementation uses modular row and column elimination. The new implementation checks every binary 3×3 incidence matrix at levels two and three. It compares the full kernel and image with the literal incidence definition, including the actual support of each multiplicity vector, and verifies every binary combination of the retained image-basis witnesses. This covers 1,024 systems, 294,912 multiplicity vectors, and 2,928 carried witnesses. The dual-lattice method also reproduces every kernel of a 2×2 matrix over $\mathbb{Z}/4\mathbb{Z}$, including 81 kernels of cardinality two or eight, which are nonfree over that ring.

Sixteen additional Boolean-pattern systems test the closure bound at levels up to $r = 256$. All agree with direct subset enumeration, and all attain the bound $B(r, t)$ for their parameters. Two deliberately incorrect variants—omitting triple seeds and omitting the factor two in T_v —give different lattices from the exact construction on the designated control instances.

A focused check of the forward-substitution formula (18) revisits the 256 general mod-four matrices and these sixteen Boolean-pattern systems, together with three unconstrained boundary cases. Every division is exact, all resulting matrices agree with qH^{-1} , and all exhaustively enumerated mod-four kernels agree. This check reuses the preceding script’s Hermite and incidence routines; it is a regression check of the explicit recurrence, not an independent implementation or an additional sample of graph problems.

7.2 Checks of the graph reduction

For connected labelled graphs of orders two through four, we enumerate independent supports and levels one through three. When the toggle space is nonzero, we compare two computations: direct enumeration of the possible toggles followed by graph search over the allowed ordinary complementations, and the auxiliary-graph construction followed by the constrained local-Clifford equations. There are 2,460 comparisons: 1,020 accept and 1,440 reject, with no disagreements. In each case we also enumerate the entire linear solution space of the Clifford equations before checking the determinant conditions. These tests exercise the restricted graph-operation reduction; arbitrary chosen supports need not be type- X sets of a standard form. The 261 zero-image support/level cases are therefore excluded from this comparison, since Lemma 6.5 relies on standard form. A further 1,105 ordinary-LC comparisons agree with unrestricted graph search on small connected graphs satisfying the even-degree hypothesis.

We also use the 27-vertex construction of Tsimakuridze and Gühne [3, Section 7 and Figure 7] to test an operation beyond ordinary LC. Let Z contain six vertices and let X contain one vertex for each four-element or five-element subset of Z , adjacent precisely to that subset. Thus $|X| = 21$. Let G be this bipartite graph, and let H add the clique on Z . Giving every vertex of X multiplicity one yields ten common neighbors for every pair in Z and six for every triple. At level two, all incidence conditions hold and every pair is toggled. Hence this operation takes G to H .

The stars at vertices of X form a common MLS cover for which both graphs are in standard form. To check this, for each of the 42 stars in the two graphs the code solves for stabilizers supported inside the star and finds only its defining nonidentity generator. The computed toggle space has dimension 15 and contains the clique, with an explicitly recovered multiplicity witness. The constrained gadget test accepts. In contrast, enumerating all 2^8 solutions of the ordinary-LC linear equations finds none satisfying the local determinant conditions, so the ordinary-LC test rejects. The graph pair and its LU-but-not-LC property are from [3]; our computations use this known example as a diagnostic control for the reduction.

All new checks passed in a network-disabled container using Python and SymPy 1.14.0.

New recheck family	Cases	Outcome
Boolean-pattern systems, $r \leq 256$	16	Exact closure and bound
Dual kernels of 2×2 matrices over $\mathbb{Z}/4\mathbb{Z}$	256	All kernels agree
Binary 3×3 incidence systems	1,024	Full kernels and images agree
Carried image-witness combinations	2,928	All valid
Constrained gadget comparisons	2,460	1,020 yes; 1,440 no
Ordinary-LC graph-search comparisons	1,105	379 yes; 726 no
27-vertex higher-level control	1	LU yes; ordinary LC no

Table 1: Exact rechecks in the revised manuscript. Rows describe different objects and should not be added into a single sample count.

The archive contains the scripts, full outputs, source locators, evidence map, and unchanged round-21 proof. These are reproducible internal checks, not external mathematical review.

8 Discussion

The improvement comes from the weighted structure of the incidence rows. Closure under coordinatewise multiplication by $2a_v$ replaces subset enumeration, while Hermite reduction keeps at most $|X|$ rows with bounded entries. The subsequent kernel and image calculations preserve every admissible operation, so the rest of the Claudet–Perdrix reduction applies without changing its target space.

The polynomial bound is a worst-case decision result. The $O(n^3)$ auxiliary graph can make a direct implementation expensive, and we have not optimized the resulting exponent. A complete implementation still needs the common-MLS-cover and standardization routines. It would also allow practical comparison with the original subset-based method, which may be faster on small instances. Arbitrary qubit permutations remain outside the labelled decision problem considered here.

Authorship, preparation, and availability

The proof candidate and its exposition were developed with substantial AI assistance under the author’s direction. Two historical internal AI reviews reported no error in the frozen round-21 proof. The revised manuscript includes the sharper closure bound (7), the dual-lattice proof of Lemma 4.3, and the rechecks in Section 7. The present expansion specifies the Hermite convention, derives an integral kernel algorithm with explicit bit bounds, and details the constrained graph reduction and its determinant test. It also supplies source-numbering locators and attributes the known 27-vertex control. The historical reviews are not reviews of these revisions. Independent expert verification, formal proof-assistant verification, and historical-priority confirmation remain pending.

Scientific Agent Skills [9] guided evidence tracking and scientific writing; it did not verify the mathematics. The frozen proof and submission are available in [QIQC issue 97](#). The [research page](#) hosts this manuscript and the reproducibility package. The frozen proof was contributed under CC BY 4.0 and the original diagnostic scripts under Apache-2.0; cited third-party works retain their own terms.

A Source versions and statement locators

The main graph-reduction source is the 27-page PDF of arXiv:2502.06566v3, linked in [4]. Its statement numbers differ from those in the HTML conversion of that same version, and from the 20-page conference proceedings. All locators in the main text use the v3 PDF. The table below provides the corresponding HTML numbers for readers following earlier links. PDF page numbers count from the first page of the article.

Result used	PDF statement	Page	HTML statement
Incidence and generalized operation	Defs. 1–2	4	Defs. 2.1–2.2
Clifford equations	Prop. 7	6–7	Prop. 3.1
Affine determinant solution set	Lemma 8	7	Lemma 3.2
Constrained decision	Prop. 10	8	Prop. 3.5
Local sets, types, standard form	Defs. 12–14	8–9	Defs. 3.8–3.10
Common standardization	Lemma 16	9	Lemma 3.12
Matching types and neighborhoods	Lemma 17	9	Lemma 3.13
One generalized operation	Lemma 18	10	Lemma 3.14
Complete incidence image	Lemma 19	10–11	Lemma 3.15
Auxiliary graph reduction	Lemma 20	11	Lemma 3.17
Auxiliary Clifford constraints	Lemma 21	11–12	Lemma 3.18
Logarithmic level suffices	Cor. 29	14	Cor. 3.29

Table 2: Statement correspondence in [4], arXiv v3. Appendix A gives the Clifford conversion, Appendix C the standardization, and Appendix D the auxiliary-graph reduction.

The integer normal-form complexity statement is Theorem 2 of Kannan–Bachem [8, p. 504]; the same page begins its full-row-rank rectangular extension. Our matrices have full column rank, so we apply that extension after transposition. The polynomial LU-decision question appears in Chapter 8, p. 107 of the v2 thesis [7]. The 27-vertex diagnostic is the graph pair described in Section 7, Figures 7–8, pp. 15–16 of arXiv:1611.06938 [3]. These locators distinguish imported results and known controls from the incidence-compression argument proved here.

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